

Dry run all the programs and submit it in the next class

a)

```
int k,j;
    cout<<"Enter the Number"<<endl;// enter 10
    cin>>k;
    int z=0,i=1;
    while (i <= k){
        z=z+i;
        i=i+1;
    }
    while (k >=1){
        j=1;
        while (j <= k){
            cout<<z-k+j<<" ";
            j=j+1;
        }
        z=z-k;
        cout<<endl;
        k=k-1;
    }
```

b)

```
int i=1;
int NoOfStars=1;
int NoOfSpace=6;
int NumberOfLines=9,j,space;
while (i<=NumberOfLines){
    j=1;
    while (j<=NoOfStars){
        if( i%2!=0)
            cout<< "*"<<" ";
        else
            cout<< " "<<"*";
        j+=1;
    }
    space=1;
    while (space<=NoOfSpace){
        cout<<" ";
        space+=1;
    }
    if (i==(NumberOfLines/2)+1)
        NoOfStars-=1;

    j=1;
    while (j<=NoOfStars){
        if (i%2==0)
            cout<<"*"<<" ";
        else
            cout<<" "<<"*";
        j+=1;
    }
    if (i==(NumberOfLines/2)+1)
        NoOfStars+=1;
    cout<<endl;

    if (i%2==0 && i<=(NumberOfLines/2)+1){
        NoOfStars+=1;
        NoOfSpace-=4;
    }
    else if (i%2!=0 && i>=(NumberOfLines/2)+1){
        NoOfStars-=1;
        NoOfSpace+=4;
    }
    i+=1;}

```

c)

int height,mid,r,c;

```

cout<<"Enter an odd integer as the height";
cin>>height;
if ((height<= 0) || ((height %2) == 0)){
    cout<<"Invalid Height"<<endl;
}
else{
    mid=height/2+1;
    r=mid;
    while (r>=1){
        c=1;
        while (c<=height){
            if(c==r || c==(height+1-r))
                cout<<'*';
            else
                cout<<' ';
            c+=1;
        }
        cout<<'\n';
        r-=1;
    }
    r=2;
    while (r<=mid){
        c=1;
        while (c<=height){
            if (c==r || c==(height+1-r))
                cout<<'*';
            else
                cout<<' ';
            c+=1;
        }
        cout<<'\n';
        r+=1;
    }
}

```

Practice Problems:

[Hint: Please use Pen & Paper to first design the Algorithm and then c++ compiler]

Q1: Given an input number, print its factorial. For example, factorial of 5 = $5 * 4 * 3 * 2 * 1 = 120$

Q2: Write a C++ program to find those numbers which are divisible by 7 and multiple of 5, between 1500 and 2700 (both included).

Q3: Write a C++ program to guess a number between 1 to 9.

Note : User is prompted to enter a guess. If the user guess wrong then the prompt appears again until the guess is correct, on successful guess user will get a "Well guessed!" message, and the program will exit.

Note: use `srand(time(0))` and `rand()` to generate guess number.

Q4: A Harshad number “is an integer that is divisible by the sum of its digits”
(Wikipedia)

Example: $81 \rightarrow 8 + 1 = 9 \rightarrow 81/9 = 9 \rightarrow \text{Harshad!}$

Note: Numbers length unknown.

Q5: Write a C++ program to count the number of even and odd numbers from a series of (positive) numbers. Program should except input until user enters -1, it should then print number of odds and evens

Sample numbers : numbers = (1, 2, 3, 4, 5, 6, 7, 8, 9, -1)

Expected Output :

Number of even numbers : 5

Number of odd numbers : 4

Q6: Write a C++ program to get the Fibonacci series between 0 to 50.

Note : The Fibonacci Sequence is the series of numbers :

0, 1, 1, 2, 3, 5, 8, 13, 21,

Every next number is found by adding up the two numbers before it.

Expected Output : 1 1 2 3 5 8 13 21 34

Q7: Write a C++ program to print the following series

1, 2, 4, 8, 16, 32, 64, ...

1st => 1

2nd => 2

... 7th => 64

So print 100th number.

Q8: Write a C++ program to print the following series

0, 1, 3, 6, 10, 15, 21, ...

1st => 0

2nd => 1

... 7th => 21

So print 100th number.

Q9: Write a C++ program to print the following series

2, 5, 7, 14, 26, 47, ...

1st => 2

2nd => 5

3rd => 7

... 6th => 47

So print 100th number.